

ANDREW YING

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SUMMARY: Highly motivated graduate with experience in surgical robotics simulation and biomedical research. Strong technical skills in programming, graphics, and machine learning. Seeking to develop medical devices to enhance surgeon performance and improve patient outcomes.

EDUCATION

Johns Hopkins University	Baltimore, MD
M.S.E. Biomedical Engineering	August 2024 - May 2025
B.S. Biomedical Engineering	August 2021 - May 2025
Dual Major in Computer Engineering	
B.S. GPA: 3.97; M.S.E. GPA: 4.00	

Honors: Hodson Trust Scholarship, John Boswell Whitehead Award, Dean's List Fall 2021 – Spring 2025

Coursework: Deep Learning; Computer Vision; Computer Integrated Surgery; Robot Devices, Kinematics, Dynamics, and Control; Computer Graphics; Intermediate Programming; Advanced Digital Systems; Signals & Systems

WORK EXPERIENCE

Laboratory for Computational Sensing + Robotics	Baltimore, MD
Graduate Researcher under Dr. Adnan Munawar	January 2025 – Present

- Developed virtual reality simulator using ROS, Chai3D, and PyBullet Physics for transsphenoidal surgery
- Implemented endoscopic camera with radial and tangential distortion using OpenGL and GLSL shaders
- Segmented drillable facial bone and flexible nasal passage in 3D Slicer for realistic soft body collisions
- Paper accepted for publication to IEEE BioRob 2026 conference

Kadoch Lab, Harvard Medical School/Dana-Farber Cancer Institute	Boston, MA
Research Intern	May 2024 – May 2025
Research Assistant	January 2026 – Present

- Analyzed RNA-seq, ATAC-seq, and CUT&RUN data to characterize the role of the BAF complex in CBFA2T3-GLIS2 fusion positive AML
- Trained random forest models to identify biomarkers associated with cancer dependency on BAF complex
- Generated Kaplan-Meier curves to investigate clinical impact of ncBAF and SAGA complex interactions

Precision Care Medicine	Baltimore, MD
Team Member under Dr. Nitish Thakor	August 2024 – May 2025

- Analyzed stroke patient gait metrics produced by pressure-sensing insole to quantify therapy effectiveness
- Presented project at Institute for Computation Medicine Night and Design Day 2025

Biomedical Engineering Design Team	Baltimore, MD
Design Team Member under Dr. Hien Nguyen	March 2022 – May 2024

- Designed AI surgical training feedback and analytics platform using Flutter to improve resident education
- Presented project at Design Day 2023 and Whiting School of Engineering Advisory Board Meeting
- Received the Catalyst Award from the Hopkins Office of Undergraduate Research

LEADERSHIP AND VOLUNTEERING

MedTech Network (MTN)	Baltimore, MD
Vice President: Workshops Committee	September 2021 – May 2025

- Marketed and organized over 20 technical workshops on CAD, 3D printing, and Python for student body
- Taught 40 8th grade students about Arduino pulse sensor and breadboards at Science Olympiad Stem Expo

Teaching Assistant	Baltimore, MD
Biomedical Data Science; Linear Signals & Systems	August 2024 – December 2024

- Led section and taught 25 students exploratory data analysis in Python using Pandas and Numpy
- Answered questions regarding machine learning using scikit-learn and PyTorch during office hours

TECHNICAL SKILLS

Software: C, C++, Python, ROS, OpenGL, SOLIDWORKS, Linux, Slurm, PyTorch, Pandas, Matplotlib, R, MATLAB

Manufacturing: 3D Printing, Manual Mill, Laser Cutter, Plasma Router